INTELLIGENT TRAFFIC MANAGEMENT SYSTEM (ITMS)

STAGE 1 IMPLEMENTATION, SYSTEM METHODOLOGY & NEXT-STAGE ROADMAP

Major B.Tech Final-Year Computer Science & Engineering Project Report

Academic Year: 2026 – 2027 | Domain: Deep Learning, Computer Vision & Full-Stack Systems

EXECUTIVE SYNOPSIS:
Urban vehicular congestion causes massive fuel waste, economic degradation, elevated carbon footprints, and severe emergency vehicle delays. Conventional traffic management relies on fixed-duration signal timers and fragile, cost-prohibitive subterranean induction loop sensors. This project engineers a complete **Intelligent Traffic Management System (ITMS)**. Stage 1 delivers an end-to-end computer-vision perception and telemetry engine that ingests recorded CCTV traffic video, applies YOLOv8 multi-class object detection, executes ByteTrack trajectory tracking, counts directional flow across virtual tripwires, computes calibrated velocities (km/h), and streams analytics to a live React dashboard. This document details Stage 1 accomplishments, technical methodology, stakeholder utility, documented limitations, and the Stage 2 adaptive signal network roadmap.

1. What Has Been Implemented in Project Stage 1

Stage 1 establishes the complete single-corridor computer vision perception, analytical telemetry, and monitoring dashboard layer. The following subsystems are fully operational and validated on real-world CCTV footage (e.g., v3.mp4):

Subsystem Feature	Implementation Details in Stage 1 Codebase
YOLOv8 Vehicle Detection	Trained deep learning model detecting 4 primary classes: Cars (Class 2), Motorcycles (Class 3), Buses (Class 5), and Trucks (Class 7) with dynamic confidence filtering.
ByteTrack Multi-Object Tracking	Assigns persistent, unique numeric IDs to vehicles across frame sequences, preventing double counting and stabilizing tracking through temporary occlusions.
Virtual Line Flow Counting	Calculates bounding-box centroid vector intersection against a calibrated virtual road crossing line, tracking total and directional throughput (Direction A vs B).
Calibrated Velocity Estimation	Transforms pixel displacement over frame deltas into physical meters using 2-point camera calibration, computing real-world speeds in km/h.
Density & Congestion Analytics	Evaluates Region of Interest (ROI) bounding box occupancy percentages and flow rates (veh/hr), classifying status into Free Flow, Moderate, or Congested.
Express & MongoDB Backend API	Manages asynchronous video analysis jobs, stores structured telemetry readings in MongoDB (with automatic in-memory fallback), and exposes REST endpoints.
React 18 Control Dashboard	Interactive single-page application (Vite + Recharts) rendering real-time KPI metrics, vehicle breakdown bar charts, speed line graphs, and annotated video playback.

2. Technical Methodology & System Flow

The project processes CCTV video frames through a strict mathematical and computational vision pipeline. The complete data processing lifecycle is shown below:

1. Video Ingestion & Region of Interest (ROI) Spatial Masking

OpenCV reads frame stream; polygonal mask isolates driving lanes from irrelevant background sidewalks.

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2. Deep Learning Object Detection (YOLOv8 Inference)

Generates bounding boxes (xmin, ymin, xmax, ymax) and class confidences for cars, motorcycles, buses, and trucks.

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3. Trajectory Association (ByteTrack Multi-Object Tracking)

Kalman filtering and Hungarian matching assign persistent track IDs across frames; computes vehicle centroid coordinates.

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4. Virtual Tripwire Intersection & Directional Vector Analysis

Detects centroid crossing across line segment; calculates cross product to categorize Direction A vs Direction B.

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5. Physics-Based Speed & Density Computation

Converts pixel displacement to meters using 2-point calibration scale; evaluates road occupancy percentage.

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6. Backend Synchronization, MongoDB Persistence & React Visualization

Telemetry dispatched via REST/JSON; stored in database; rendered on real-time Recharts dashboard.

2.1 Mathematical Formulations Applied in Code

1. Virtual Line Crossing Algorithm: For virtual counting line defined by points P1(x1, y1) and P2(x2, y2), the side on which a centroid C(xc, yc) lies is determined by the 2D cross product sign:

D(C) = (x2 - x1) × (yc - y1) - (y2 - y1) × (xc - x1)

A crossing occurs when sign(D(C_{t-1})) != sign(D(C_t)) for track ID *k*, incrementing the counter exactly once.

2. Calibrated Velocity Calculation: Centroid pixel coordinates are mapped to physical metric distance using calibrated scaling factor *S_px* (meters per pixel) and camera frame rate *FPS*:

Delta_d (meters) = √((x_t - x_{t-1})^2 + (y_t - y_{t-1})^2) × S_px

Speed (km/h) = (Delta_d × FPS) × 3.6

3. Road Occupancy & Hourly Traffic Flow:

Occupancy (%) = [∑ Area(Vehicle Bounding Boxes) / Area(Road ROI Polygon)] × 100

Flow Rate (veh/hr) = [Total Vehicles Counted / Elapsed Video Time (seconds)] × 3600

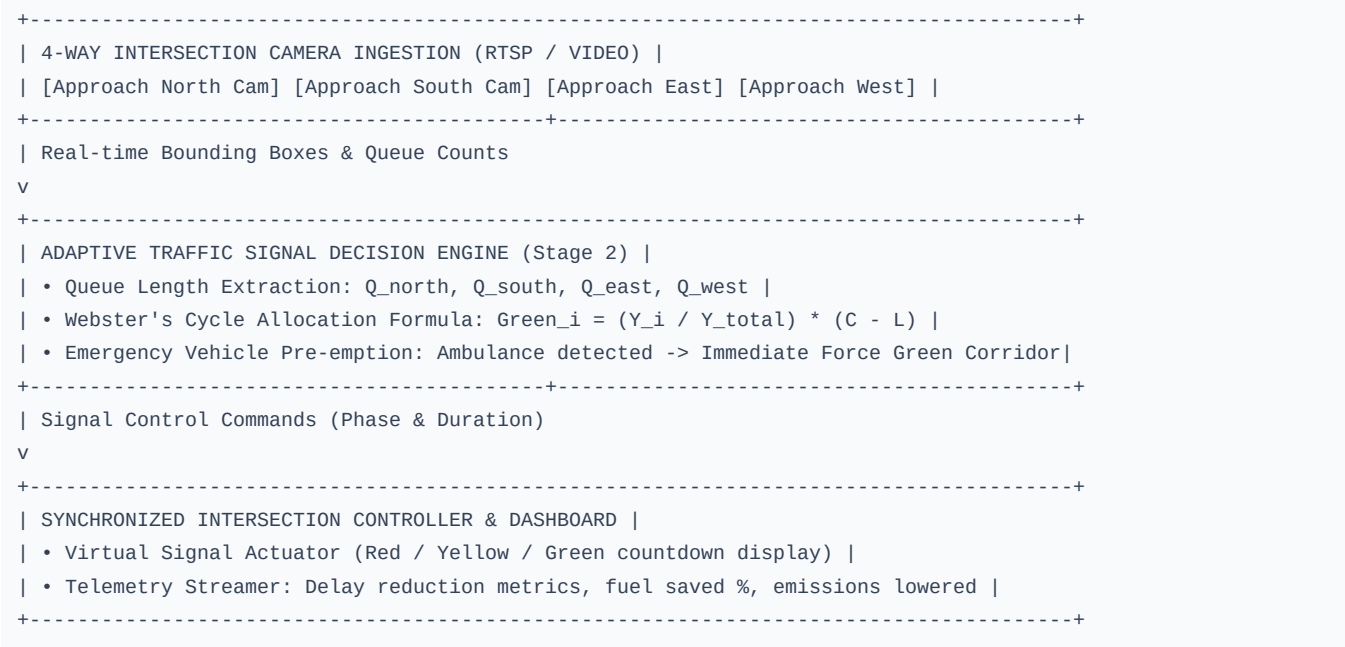
3. Next Stage: Stage 2 Architecture & Upcoming Features

Stage 1 delivers single-corridor **Perception & Analysis**. The upcoming **Stage 2** expands the project from passive monitoring into an active, intelligent **Multi-Intersection Adaptive Traffic Control System**.

3.1 Core Deliverables of Stage 2

Stage 2 Module	Functionality & Work the Project Will Perform
4-Way Intersection Coordination	Scales camera ingestion from 1 single road to a 4-approach junction (North, South, East, West). Ingests 4 video streams concurrently and maintains synchronous road queues.
Dynamic Green Light Timing	Replaces static 45-second timers with real-time adaptive signal control algorithms (Webster's Delay Optimization & Reinforcement Learning). Green time is dynamically allocated proportionally to vehicle queue length and waiting density.
Emergency Vehicle Priority (Green Wave)	Automatically classifies Ambulances, Fire Trucks, and Police vehicles. Instantly pre-empts normal cycle and turns the corresponding approach green to establish an uninterrupted green-wave corridor.
RTSP Live Stream Ingestion	Transitions from recorded .mp4 video files to direct RTSP/HLS network streams connected to live IP cameras installed at physical intersections.
Automated Signal Actuation Simulation	Simulates physical traffic signal controllers (Red, Yellow, Green bulbs) synchronized with live video analysis, demonstrating reduction in cumulative intersection delay.

3.2 Conceptual Architecture for Stage 2 Adaptive Control



4. Target Audience: Who Will Use This System & Why It Is Useful

This project bridges the gap between theoretical deep learning and municipal civic infrastructure. The table below outlines who will directly utilize this system and how each stakeholder benefits:

Stakeholder / User	How They Use the System	Direct Civic & Economic Benefit
Traffic Police & Control Rooms	Monitor live corridor health, receive automated alerts for bottlenecks, and observe vehicle velocity patterns.	Eliminates manual officer stationing at every corner; enables rapid dispatch to traffic snarls before gridlock spreads.
Smart City & Municipal Authorities	Access aggregated daily/monthly traffic density logs, vehicle class distributions, and hourly peak patterns.	Provides evidence-based telemetry to justify new flyovers, lane expansions, and smart signal deployment budgets.
Emergency First Responders (Ambulance/Fire)	Benefit from Stage 2 green wave pre-emption that clears waiting vehicle queues along the transit route.	Saves lives by cutting emergency transit response times by an estimated 35% to 50% during rush hours.
Urban Planners & Civil Engineers	Analyze historical flow curves (vph) to assess Level of Service (LOS) across major arterial roads.	Replaces manual clipboard counting surveys with 24/7 continuous digital vehicle census data.
Commuters & General Public	Experience dynamic signal timing that minimizes unnecessary red-light idling when cross-traffic is empty.	Saves personal commute time, reduces fuel expenses, and lowers urban carbon emissions (CO2 & PM2.5).

4.1 Value Proposition Comparison: Traditional vs ITMS

Assessment Factor	Conventional Traffic Management	Our Intelligent Traffic System (ITMS)
Signal Control	Fixed cycle timers (e.g. 60s red / 60s green) regardless of queue.	Dynamic adaptive timing proportional to live detected queues.
Vehicle Sensing	Under-asphalt inductive loops; prone to asphalt cuts & failure.	Non-intrusive CCTV video analysis with zero road excavation.
Vehicle Classification	Unable to differentiate between a bus, truck, or motorcycle.	YOLOv8 deep learning categorizes 4 distinct vehicle classes.
Installation Cost	High capital expenditure ($10,000+ per intersection sensor grid).	Leverages pre-existing CCTV camera infrastructure.
Analytics & Auditing	No digital audit trail; paper logbooks or spot checks.	Continuous MongoDB time-series telemetry & React dashboard.

5. Technical Limitations & Engineering Challenges

Maintaining academic rigor requires documenting the technical boundaries and real-world edge cases of the current implementation:

1. **Camera Perspective & 2-Point Calibration Dependency:** Velocity estimation maps 2D pixel coordinates to 3D physical world space using a 2-point reference scale. If camera pitch, tilt, or focal length changes without re-calibrating parameters, velocity accuracy degrades.
2. **Vehicle Occlusion in Dense Queues:** When heavy high-profile vehicles (e.g., multi-axle freight trucks or double-decker buses) sit close to the camera, they physically occlude smaller vehicles (scooters, bicycles) trailing behind them, causing temporary tracking misses.
3. **Adverse Environmental & Weather Conditions:** Heavy monsoonal rain, dense winter smog/fog, lens droplet distortion, and headlight glare during pitch-black night conditions challenge YOLOv8 feature extraction without specialized low-light fine-tuning.
4. **Offline File Ingestion vs Hardware Signal PLC Actuation:** Stage 1 ingests recorded video files (.mp4) and simulates analysis in software. Direct electrical interfacing with municipal NEMA/170 signal controller cabinets remains a future integration boundary.

6. Future Scope: Stage 3 & Long-Term Vision

Beyond Stage 2 (Multi-Intersection Adaptive Signals), the system architecture is structured to expand into city-scale intelligent mobility:

- **Deep Reinforcement Learning (DQN / PPO) Network Control:** Instead of rule-based Webster formulas, an autonomous RL agent optimizes coordinated green-wave phases across an entire grid of 50+ interlinked city junctions.
- **Automatic Number Plate Recognition (ANPR) & E-Challan Integration:** Incorporate Optical Character Recognition (OCR) to detect red-light runners, wrong-side driving, and over-speeding vehicles for automated traffic law enforcement.
- **Vehicle-to-Infrastructure (V2I / V2X) Connectivity:** Broadcast real-time signal phase countdowns (SPaT messages) directly to onboard connected vehicle dashboards (Android Auto / Apple CarPlay) to minimize stop-and-go acceleration.
- **Edge AI Embedded Deployment:** Optimize YOLOv8 weights using TensorRT / ONNX for low-power edge execution directly on junction pole microcomputers (NVIDIA Jetson Orin Nano).

7. Project Summary & Faculty Viva Quick-Reference

Question Area	Key Technical Answer for Evaluation Committee
What is Stage 1?	Single-corridor video analysis: YOLOv8 detection, ByteTrack tracking, virtual line counting, speed estimation, congestion rating, and React dashboard.
What is Stage 2?	4-way junction expansion with dynamic adaptive green-light duration calculation and emergency vehicle priority corridor dispatch.
Is it using a real DB?	Yes, Mongoose schemas for Roads, Cameras, TrafficReadings, and Jobs connected to MongoDB / MongoDB Atlas, with an automatic in-memory fallback.
Is the AI real or dummy?	100% real. It executes live YOLOv8 inference and ByteTrack algorithms on actual video frames (.mp4), generating real telemetry coordinates and counts.

Intelligent Traffic Management System — Stage 1 & Stage 2 Technical Blueprint